

Lecture 2: Option Pricing & A Two-State Tree Model

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Based on lecture notes by me and many previous CS335 instructors

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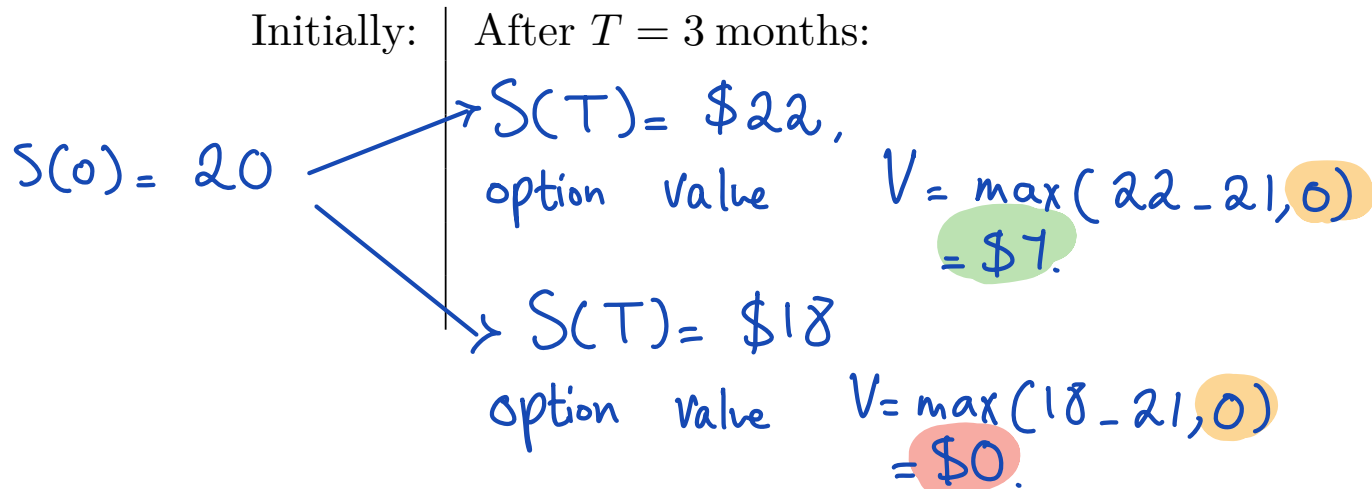
A Simplified Scenario - Two-State Tree

Suppose the current stock price at $(t = 0)$ is $S = \$20$.

After 3 months, let's assume the stock price will become one of ...
$$S = \begin{cases} \$22 \\ \$18 \end{cases}$$

We wish to price (determine the value of) a European Call option with strike price $K = \$21$, and expiry $T = 3$ months.

States at Expiry T



Now, suppose we somehow know the probability of each outcome:

$$\text{prob}(S \rightarrow 22) = P_{\text{up}} = 0.1$$
$$\text{prob}(S \rightarrow 18) = P_{\text{dn}} = 0.9$$

How can we determine how much the option is worth **today** (at time $t = 0$)?

“Obvious” (naïve) pricing method

“Expected discounted cash flow”:

The expected value of the option (in the statistical sense) at $T=3$ is:

$$\sum p_i V_i = 0.1 \times 1 + 0.9 \times 0 = 0.1$$

But this is the (future) value at $T = 3$. We ought to discount this cash flow back to the present, $t = 0$, to reflect interest over that period.

Assume interest rate $r = 12\%$ (annually).

Then ¹

$$\text{The present value} = e^{-(0.12 \times \frac{3}{12})} \times 0.1 \approx \$0.097$$

e^{-rt} :

- $r = 0.12$ (12% annual rate)
- $t = \frac{3}{12} = 0.25$ years (3 months)
- The future expected payoff = 0.1

¹If you forget what continuously compounding interest rates are:

Thought Experiment

Suppose a *tricky/sneaky/greedy banker* offers to buy an option from you for \$0.30.

This is \$0.203 more than what we just worked out the option value to be (according to expected discounted cash flow).

Should you accept this *generous offer*? Is what we considered a wise way to price an option?

Thought Experiment

According to the preceding logic, we should *expect on average* a profit of \$0.203 for each option! Woohoo!

Sneaky!

Not so fast! There exists a strategy that will **guarantee** the *greedy banker* a profit of \$0.33 for each option they purchase at \$0.30.

Note:

- 1 This is not an expected (average) profit - it is absolutely certain!
- 2 This is completely independent of changes in stock price.

How? What's the magic?

Understanding Arbitrage

What's the magic secret? First, we'll need some more definitions...

Definition: An **arbitrage opportunity** exists if, starting with 0 initial wealth, an investor can devise a strategy that ensures:

1. probability of loss $= 0$.
2. probability of gain > 0 .

Arbitrage Pricing

Instead of using expected value to price the option, consider the following strategy:

- Assemble a portfolio Π , consisting of the stock and the option.
- We'll design the portfolio such that there is no uncertainty (no risk) at all in the portfolio's value after 3 months. [How?]

Since such a portfolio is risk-free, it must earn the "risk-free interest rate". [why?]

Risk-Free Rate

Definition: The risk-free interest rate / rate of return over a period T is the rate of interest received by an investment that has zero probability of loss.

Example: Short-term government bonds.

We claimed that our portfolio Π must earn the risk-free rate since we will design to have no uncertainty *whatsoever*.

Why does a portfolio with a *certain* (guaranteed) cash flow at time T have to earn the risk-free rate?

Typical Financial Argument (“No Arbitrage”)

We'll argue that the portfolio's return must match the risk-free rate using a “proof” by contradiction...

Case 1: portfolio's return $>$ risk-free rate

This would create an *arbitrage opportunity*, according to the following strategy:

- a. Borrow cash from the bank at the risk-free rate.
- b. Construct the portfolio using that cash.
- c. After 3 months, sell the portfolio, repay the bank loan, pocket the difference (which is a profit, since the portfolio beat the risk free rate!).

An Arbitrage Opportunity

This is an arbitrage opportunity! What would happen if such an opportunity existed?

Many people would jump on it, and try to carry out that strategy. But then...

- a portfolio would be in demand
- b its price would quickly rise
- c portfolio's return would decrease, until it matches the risk-free rate again!

End of Lecture 2

Shorting the Portfolio

Case 2: Portfolio's Π 's return $<$ risk-free rate.

i.e., what is our risk-free portfolio were instead to get a lower rate of return than the risk-free rate?

Idea:

Definition:

Another Arbitrage Opportunity

This leads to a new strategy:

a

b

Another Arbitrage Opportunity

What happens if this arbitrage opportunity/strategy exists?

Again:



No Arbitrage Principle:

(We will assume this throughout the course.)